Report for lesson 4

In lesson 4 we have learnt several things, specifically from previous lectures we learnt how to deal with database, we have learnt to get database from GitHub/Kaggle change them or adapt this csv file for our needs. Also we have learnt conditions and printout input features from previous lesson. In this report I am going to show data preprocessing methods, Machine Learning projects sequence, also I am going to represent methods that help fill up empty space's in database. Also various libraries and their uses.

Lets remember few interesting features from previous lecture. the most important part in the beginning of the machine learning process is picking up and collection correct and suitable information, in other words it means Data Collection(we can do it via collecting information through internet, which Is scraping or finding already processed datasets)

Next step is Data preprocessing and this is important part because without fully processed dataset progress of the work won't change at all. Sequence of the ML project simply look like this: Data Preprocessing → Model Selection → Model Training → Model Prediction → Model Evaluation → model testing. That is why processing of the data is important, because neither algorith nor dataset will not work properly and it will cause errors during process with data.

From lecture 4 I acknowledged that empty space in dataset could be solved easily based on what type of information is dropped and what kind of information we need to fill. For example lets imagine situation where we have student at class, and we have their student ids but right now we don't know who is attending and who is not, in order to figure it out we can call them one by one and ask id number. By that we would know exact numbers of the students and we would know who attending and who is missing because list of the student has to begin for example 1-20 and there would definitely some missing numbers between them if some student wont be at the class. It is not hard to figure it our right, but it is only this easy cause we know data type but imagine that we don't know what kind of information we have to fill, this is exactly what we have to dealt with when we trying to fill missing infromation.

Therefore we have 5 methods in data preprocessing thatwill help us to fill them up as much accurately as possible. They are Mean method(which can be used only in numerical rows and it basically the way to find out average arithmetic missing number), second one is Mode method(it is a method that can be used both in numberical and categorical ways, because it's target is to find out most repeatable answer and give it to us). Third one is Median method, it can be understood that method is just pick for us the most median answer from all dataset and fill out rest empty holes with this information. Next one is fixed method, this is unique method due to It not uses any specefic algorith to pick number instead we have to pick information (numerical or categorical) that has to correct and 100%

reliable and all of the empty holes in dataset will be replaced by this value. Last one is drop method which basically drops whole row of the information if we target void feature in the dataset, that is why we have to use this method carefully, because it can deal damage to our dataset.

Additionally, we can talk about libraries, what even library means in term of programming?. My personal answer Is massive block of code that consist of modules, objects, and lots of packaged code that can easily archived to pc and used for person’s needs. There are a lot of libraries that currently millions of developers uses on a daily basis. Simple example: Pandas, numpy most useful library in my view for optimazation even though I heard just a little that can this lib capaple of.